

Directions for use
Read carefully!

Compound Sodium Lactate Intravenous Infusion B.P. (Hartmann's Solution) Half-Strength, with 2.5% Glucose

Isotonic electrolyte solution with free water.

Composition

Each 100 ml contains:

Sodium Chloride	0.300 g
Sodium Lactate	0.157 g
Potassium Chloride	0.020 g
Calcium Chloride·2H ₂ O	0.014 g
Glucose Monohydrate for Parenteral Use	2.750 g
Water for injections to	100 ml

Electrolytes:

Na ⁺	mmol/l
K ⁺	65
Ca ⁺⁺	2.5
Cl ⁻	1
Bicarbonate ⁻ (as lactate ⁻)	56
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Caloric value:

425 kJ/l ± 100 kcal/l

Osmolarity:

278 mOsm/l

Characteristics

Compound Sodium Lactate Intravenous Infusion half-strength contains only half of the original full electrolyte concentration of the **Compound Sodium Lactate Intravenous Infusion** (Hartmann's Solution) and supplies physiological free water. The lactate ion is gradually metabolised and converted to bicarbonate in a concentration half of that of the extracellular fluid. Additionally the lactate ion exerts a slightly alkalinising effect.

Mechanisms of action

Sodium Chloride is the principal salt involved in maintaining the osmotic tension of the blood and tissue. Changes in sodium and chloride levels change this osmotic tension and hence influence the movement of fluids and diffusion of salts in cellular tissue.

Sodium Lactate after absorption, is metabolised in 1 to 2 hours to bicarbonate, it then behaves as endogenous, bicarbonate, exerting an alkalinizing effect. In the absence of bicarbonate deficiency, it is excreted by the kidney; urine becomes less acidic with accompanying diuresis.

Potassium Chloride provides potassium ions to the body. Potassium is the principle cation of intracellular fluid and is intimately involved in cell function and metabolism. It is essential for carbohydrate metabolism, glycogen storage and for protein synthesis. Like Sodium, it is integral in maintaining transmembrane potential and profoundly affects muscle, including the myocardium.

Similarly, Calcium Chloride provides calcium ions. Calcium is involved in the maintenance of normal muscle and nerve function, normal cardiac function as well as normal blood clotting.

Glucose is absorbed from the gastrointestinal tract, and oxidised as a source of energy, or stored in the liver as glycogen. It is the only energy substrate which is directly, instantly and universally utilized by the body. Glucose is irreplaceable and vital for the same myocardium, brain and nerves.

